

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation	Predicts the most suitable crop using soil and environmental parameters.	Prediction Module	Done	High
2	Data Preprocessing	Cleans the dataset, handles outliers, and prepares data for model training.	Preprocessing Module	Done	Medium
3	Exploratory Data Analysis (EDA)	Performs univariate, bivariate, and multivariate analysis of the dataset.	Data Analysis Module	Done	Medium
4	Machine Learning Model Training	Machine Learning Model Training	Model Building Module	Done	High
5	Input Validation	Validates all user inputs using client-side and server-side validation.	Validation Module	Done	Medium
6	Responsive Web Interface	Allows users to enter soil and weather parameters through a simple web interface.	Frontend Module	Done	High
7	Prediction Result Display	Displays the recommended crop based on user inputs.	Flask Application	Done	High
8	Model Integration	Integrates the trained Machine Learning model with the Flask application for real-time prediction.	Backend Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	10

Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Crop Recommendation Form
2	Backend / Logic	3	Crop Prediction Engine, Model Integration, Input Validation
3	Database / Storage	1	Crop Recommendation Dataset and Trained Model (Joblib)
4	API / Integration	1	Flask Routing and Machine Learning Model Integration
5	Security / Authentication	1	Client-side and Server-side Input Validation